Learning to Use What a Model Knows

Self-generated supervision for compositional factual reasoning

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The question

Can a language model improve how it combines facts using supervision generated by the same model, without external answer labels in its training targets? The project studies two-hop factual questions and follows the evidence from measuring failures to controlled training, transfer, and mechanism diagnosis.

The answer so far

There is a reproducible but modest benefit on the original task. Context-conditioned teacher answers outperform direct self-distillation even after approximate matching of target accuracy. This is evidence for the value of how self-generated targets are constructed; it does not yet identify the internal mechanism or establish general multi-hop improvement.

Finding	Principal result	Interpretation
Compositional failure persists	33.80% failure among 2,379 Qwen3-14B items with both hops correct	Operational knowledge does not guarantee composition.
Self-target training helps	C accuracy: 49.60% to 53.13%; +3.53 pp	Three-seed, within-dataset double holdout.
Teacher context matters	Matched exp12: +2.48 pp over direct targets; 95% interval [1.83, 3.12]	Six seeds; fixed 303-item reused evaluation subset.
Broad transfer is unconfirmed	MuSiQue two-hop EM: +0.217 pp; [-0.087, 0.521]	Predefined transfer criterion not met.

Sources: E1, E8-9, E12m, E13. pp = percentage points. Intervals here describe training-seed variation on fixed items, not population-wide uncertainty. Source key on page 9.

Research contribution at this stage: an empirical account of when self-generated supervision helps, which simple explanations are weakened by controls, and where the effect fails to generalize. The broader claim of autonomous self-improvement remains open.

Reading guide: setup and measurement (p. 2); training controls (pp. 3-4); teacher-source comparison (p. 5); external transfer (p. 6); current diagnosis (p. 7); historical evidence and conclusions (p. 8); sources (p. 9).

1. From factual knowledge to composition

Task and training method

The original benchmark is Compositional Celebrities [1]. A question combines a person-to-birth-country fact with a country attribute, such as its capital. The compositionality gap is the proportion of composed questions answered incorrectly *among items whose two constituent questions are both answered correctly*. "Knowing" is therefore an operational property under a fixed prompt and decoding setup.

Self/context teacher: answer hop 1 → place the model's own answer in a background statement → answer the composed question → train a student to answer the original question without that background.
Direct teacher control: generate the target directly from the composed question. Both use the same base model; neither receives the gold intermediate fact when generating self targets.

Training uses Qwen3-14B with LoRA rank 16, alpha 32, dropout 0.05, AdamW at 10^{-4}, effective batch 8, and two epochs per fit unless stated otherwise. Targets are short answers; prompt tokens are masked from the loss. Generation is greedy, capped at 24 new tokens, without explicit long chain-of-thought. The original full pool is 864 questions from 216 training people. [E8-12]

Training relation families: capital, calling code, and coordinates. Held-out families: currency, top-level domain, and numeric country code. A holds out people; B holds out relation families; C holds out both (500 questions each). Countries and country facts can overlap across splits. Exp12 adds a separately reserved D set of 500 question identities; this remains a within-dataset evaluation.

Exp1 and exp7: a real gap, with measurement caveats

Model	Both hops correct (n)	Direct gap	Gold-context gap
Qwen3-0.6B	143	42.66%	60.14%
Qwen3-1.7B	947	74.66%	65.36%
Qwen3-8B	1,902	39.54%	12.83%
Qwen3-14B	2,379	33.80%	10.93%

Revised signed-floor scoring on the clean birthplace task pool (4,207 questions). Each row has a different known-item subset. Gold-context evaluation supplies the reference intermediate fact; this is distinct from the self-generated background used in training. [E1]

Fair scale comparison: on the same 738 items known by 1.7B, 8B and 14B, direct gaps are 74.12%, 26.83% and 15.45%. Scale clearly helps on this shared subset. A model-assisted audit of 120 sampled failures found 94 substantive errors, 12 reference ambiguities, 8 format cases and 6 scoring errors; it was not an independent blind human audit. [E1, E7]

Exp2: readable knowledge is not necessarily usable knowledge

After excluding intermediate/final-answer first-token overlap, an intermediate entity's first token appears in a layer's top 10 for 243/288 failures (84.38%) and 254/278 successes (91.37%). This is a readout diagnostic, not proof of full-entity representation, causal use, or a unique "too-late" failure mechanism. [E2]

2. What makes self-training useful?

Exp6 → exp8: from an initial signal to multi-seed evidence

Early self, gold-label and hop-only training motivated the later controlled series. Those early contrasts remain historical evidence (p. 8); the main quantitative training claims below use verified exp8-10 records.

Evaluation split	Base	Full self, mean ± SD	Gain
A: new people	39.00%	47.87% ± 0.50	+8.87 pp
B: new relation families	44.60%	50.80% ± 1.60	+6.20 pp
C: both new	49.60%	53.13% ± 0.90	+3.53 pp

Three training seeds; 500 items per split; 864 training examples. C seeds: 53.2%, 52.2%, 54.0%; gain interval [1.29, 5.77] pp. Exp8 and exp9 full-self summaries are the same evidence, not two independent replications. ± denotes sample standard deviation, not a confidence interval. [E8-9]

Exp8: cleaner labels do not guarantee better transfer

Target selection	Training n	C accuracy, mean ± SD
Full self	864	53.13% ± 0.90
Global confidence, top 50%	432	53.47% ± 0.95
Global random, 50%	432	54.07% ± 0.50
Global confidence, top 70%	605	52.80% ± 0.72
Global random, 70%	605	54.53% ± 0.31

For seed 0, top-half confidence filtering raises target accuracy from 51.50% to 70.60%, but it does not outperform equally sized random selection on C. Filtering also changes category composition, motivating exp9. [E8]

Exp9: question-target correspondence matters

Within-category confidence and random half-sampling give 52.53% ± 1.79 and 52.47% ± 0.90 on C. In contrast, shuffling full-pool targets within category collapses accuracy to 5.20% ± 1.51, versus 53.13% for full self. Preserving the answer distribution alone is insufficient; the pairing matters. This severe mismatch control does not isolate a specific neural mechanism. [E9]

Exp10: even targets scored wrong can yield a small gain

Category-matched condition	Training n	C accuracy	Gain over base
Own wrong targets	228	51.53% ± 0.42	+1.93 pp
Own correct targets	228	51.80% ± 0.69	+2.20 pp
Shuffled wrong targets	228	8.73% ± 0.31	-40.87 pp

Wrong minus correct: -0.27 pp, 95% seed interval [-1.52, 0.98]. This is not an equivalence result. Correct/wrong groups match category counts but use different questions; gold labels are used to select these diagnostic subsets. Automatic "wrong" is not independently adjudicated factual wrongness. [E10]

3. Can the gain compound through iteration?

Exp11: three routes, three seeds, a bounded checkpoint

The same 864 questions are reused across rounds. **Reset** trains from the base model on refreshed targets; **continue** inherits the adapter but resets optimizer and schedule; **frozen** trains on initial targets for matched cumulative gradient steps. Shared round 1 plus six later conditions produces 21 fits. This compares target refresh against more training, while generation cost and optimizer schedules are not fully matched. [E11]

Condition	C accuracy, mean ± SD	Hop-2 accuracy, mean
Base	50.00%	78.40%
Shared round 1	52.53% ± 1.33	77.07%
Reset round 2	52.93% ± 1.10	76.67%
Reset round 3	53.53% ± 0.50	76.60%
Continue round 2	53.67% ± 0.95	76.33%
Continue round 3	53.67% ± 1.01	75.60%
Frozen targets, 2-round budget	54.00% ± 1.06	77.53%
Frozen targets, 3-round budget	52.73% ± 1.36	75.00%

All metrics use the same 500 C questions. The base is 50.0%, not exp8's 49.6%: regenerating base predictions changed 18 answer strings, with four wrong-to-right and two right-to-wrong flips. Gains must use each run's own baseline. [E11]

The relevant comparison is against the budget control

Paired C contrast	Difference	95% seed interval
Reset R3 minus R1	+1.00 pp	[-3.55, 5.55]
Reset R3 minus frozen 3-round	+0.80 pp	[-3.75, 5.35]
Continue R3 minus frozen 3-round	+0.93 pp	[-3.05, 4.92]

No test in the 54-comparison primary C family survives Holm correction at 0.05. The intervals are wide with three seeds; these results do not establish either an iterative benefit or equivalence. Frozen targets at the two-round budget have the highest observed C mean, but selecting this condition on the test results would be post hoc.

More training also has costs

Hop-2 accuracy falls under both refreshed and frozen-target routes. Generated-target accuracy stays around 50-51%, while the model becomes more confident in its scored-wrong training targets. This is compatible with self-reinforcement, but changing error sets prevent a simple calibration claim. The stage took about 5.34 hours across recorded generation, loading, training and evaluation phases. [E11]

Checkpoint decision: retain the modest one-stage benefit; do not claim that repeated target refresh produces reliable compounding improvement. The next question became why one source of self-generated targets is better than another.

4. Does the teacher's context make a difference?

Exp12: direct self-distillation versus context-conditioned targets

Both arms train on the same composed questions with paired LoRA initialization and matched update counts. Context uses the model's own first-hop answer, not gold context. The original three seeds were extended to six; then existing predictions were rescored and a separate matched-target training stage was run. These are distinct analyses, not interchangeable estimates. [E12a-c]

Stage / evaluation	Context	Direct	Difference; 95% seed interval
864 training items; D500; original scorer	51.80%	49.27%	+2.53 pp [0.71, 4.36]
Same predictions; D500; revised v3 scorer	50.60%	45.70%	+4.90 pp [3.28, 6.52]
Same 864 training; fixed D303 subset; v3	See contrast	See contrast	+3.14 pp [2.53, 3.75]
New matched785 training; fixed D303; v3	71.07%	68.59%	+2.48 pp [1.83, 3.12]

All differences are context minus direct; unmatched D303 is reported as a contrast only. v3 is a post hoc, conservative whole-answer scorer; review-flagged outputs count as wrong in the main analysis. [E12b-c, E12m]

Why the scoring audit matters

The original scorer could accept substrings and incompatible answers. Under v3, 109-153 of the 500 outputs per adapted model are flagged for review. On the original six-seed predictions, jointly excluding either arm's review-flagged items gives +3.06 pp [2.46, 3.66], with 320-364 items per seed. The fixed 303-item subset excludes review flags across all 13 original models. It was selected after observing outputs, then frozen for matched785; it is not an untouched test set. [E12c]

Matched785: the strongest local control so far

The two target sets have identical original-score correct counts (359/785 each); revised accepted counts are 360 versus 359. Each of 12 fits uses 785 items, two epochs and 198 updates. On fixed D303, the base scores 67.66%; context gains +3.41 pp [3.05, 3.77] and direct +0.94 pp [0.53, 1.34]. All six paired context-minus-direct differences are positive (1.65-3.30 pp). [E12m]

Sensitivity checks remain positive: review flags counted correct yield +2.81 pp; excluding the paired union yields +2.60 pp. On full D500, the main difference is +4.30 pp [3.37, 5.23]. These scenarios are not bounds over every possible factual adjudication.

What this weakens: the simple explanation that context helps only because its targets have higher aggregate automatic accuracy.
What remains: 45 training items have review flags in at least one arm; target-token budgets differ by 6.27%; semantic error types and conditional likelihood differ. Approximate score matching is not exact factual-quality matching, and the result does not prove a knowledge-retrieval mechanism.

5. Does the effect transfer to a new dataset?

Exp13: a fixed external test, without further training

The base model and all 12 matched785 adapters were evaluated on MuSiQue [4]: 1,229 two-hop, 751 three-hop and 405 four-hop questions. This experiment uses **closed-book questions without supporting passages**; its scores are not directly comparable to the official reading-comprehension setting. The primary endpoint was two-hop exact match (EM), using the frozen alias-aware scoring protocol. [E13]

Two-hop primary endpoint (n = 1,229)	EM
Base model	6.428%
Context-trained adapters, six-seed mean	6.564%
Direct-trained adapters, six-seed mean	6.347%

Paired contrast	Difference	95% seed interval
Context minus direct	+0.217 pp	[-0.087, 0.521]
Context minus base	+0.136 pp	[-0.116, 0.387]
Direct minus base	-0.081 pp	[-0.243, 0.081]

The predeclared success rule required both context-minus-direct and context-minus-base lower interval bounds to exceed zero. Neither does. The mean context advantage equals about 2.67 additional correct answers over direct and 1.67 over base per 1,229 items. This is a small point estimate with uncertainty, not proof of zero effect. [E13]

A small, predefined known-facts subset shows a signal

Only 73/1,229 two-hop questions have both constituent answers correct under the base model. On these fixed items, EM is 24.66% for base, 26.71% for context and 24.43% for direct: context minus direct is +2.28 pp [1.54, 3.03]. However, across all six seeds, the repaired baseline failures involve only **three distinct questions**. Repeating seeds does not create more independent examples.

The subset is informative about where improvement may occur, but cannot replace the full primary endpoint. Limited measured single-hop knowledge, question reformulation and answer-reference ambiguity may all contribute to the low overall scores; none cancels the observed primary result.

Secondary endpoints do not give one uniform story

Context-minus-direct EM is -0.222 pp on three-hop questions and +0.123 pp on four-hop questions. Across all 2,385 questions, EM changes by +0.063 pp [-0.091, 0.217], while token F1 changes by +0.233 pp [0.122, 0.343]. These secondary analyses are not multiplicity-adjusted; exact correctness and token overlap should not be conflated. [E13]

Decision: preserve MuSiQue as a fixed, already-inspected transfer benchmark. The current evidence supports a local original-task effect, while overall cross-dataset transfer remains unsupported under the prespecified criterion. No tuning or rerunning of this benchmark was performed for this summary.

6. Current diagnosis: consistency is a hypothesis

Why look beyond target correctness?

Exp10 found small gains from targets scored wrong; exp12 found a context advantage after approximate target-accuracy matching. A plausible hypothesis is that a target can be factually wrong yet remain compatible with the model's own intermediate answer. An offline analysis examined existing outputs only; it did not train or query models again. [D16]

Of the matched 785 questions, 381 have identical target strings in both arms and 404 differ. On the 404 differing-target questions, context has 316 targets scored wrong (including review flags) and direct has 317. "Consistency" was tested against a dataset-derived country-attribute table and against saved model single-hop answers for the model's predicted country.

Consistency definition	Context wrong targets	Direct wrong targets	Difference
Country-attribute table	75/316 = 23.73%	6/317 = 1.89%	+21.84 pp
Model single-hop, same run	108/316 = 34.18%	16/317 = 5.05%	+29.13 pp
Model single-hop, older cache fills gaps	142/316 = 44.94%	25/317 = 7.89%	+37.05 pp

These are descriptive rates from one cached target pool, not six independent teacher samples. Unknown cases remain in the denominators. The primary attribute-table analysis has 36.71% and 41.01% unknown; arbitrary assignments to unknowns allow a difference from -19.17 to +58.55 pp. Older-cache evidence has better coverage but mixes runs. [D16]

The two wrong-target sets differ. Restricting to the same 236 questions where both targets are wrong without a review flag gives table-compatible counts 42 versus 5, a +15.68 pp difference; same-run model consistency gives 74 versus 14, or +25.42 pp. These checks support a candidate association, not a mediation estimate of the downstream training gain.

What is completed, running, and only planned

Workstream	Status at cutoff	How it enters the conclusions
Exp14: Llama-3.1-8B	User reports run in progress; no completed results found locally	No cross-family replication claim yet.
Exp15 / exp16	No separate result artifacts located under these identifiers	No missing numbers inferred; offline diagnosis is D16.
Exp17: consistency intervention	Design, feasibility counts and data-preparation artifacts only	No causal outcome; no new HPC run launched here.

The proposed exp17 control keeps the same target-string multiset and changes only its assignment to questions, under category and eligibility constraints. Existing same-run caches support 124 questions per arm under strict matching; larger-pool counts remain projections. This is intended to reduce source, length and format confounds before testing consistency causally. [P17]

Current priority: consolidate scoring/reference errors, factual coverage and composition failures; then make a bounded decision about one targeted repair. Preserve the frozen exp14 protocol and the already-inspected external benchmark.

7. Historical evidence and defensible conclusions

Exp3-6b: retain the history, qualify the evidence

The October 3 handoff records the following early results. The later exp1-11 report could not locate the corresponding original outputs for several interventions and early controls. They explain the research path, but are **not treated as reverified quantitative evidence** or used to establish a causal mechanism here. [H, R11]

Experiment	Historical result	Evidence limit
Exp3: per-item best-layer backpatching	37% repair versus 24% control	Exploratory layer selection can inflate effects.
Exp4: fixed-layer backpatching	14-16% repair versus 3% control; 5-9% damage to correct items	Reported negative net utility; denominators and causal inference need raw-data verification.
Exp5: confidence-gated patching	AUROC about 0.75; net gain about zero	Historical summary only.
Exp6: self / gold / hop-only	C base 49.6%; self 54.2% and 52.2%; gold 56.8%; hop-only 49.8%	Early seeds and controls are not substituted for later verified runs.
Exp6b: post-training lens	Self reportedly shifts intermediate readout slightly later; gold earlier	Does not establish that training repairs a timing bottleneck.

What the completed evidence supports

1. A measurable composition problem. Conditional failures remain after revised scoring. Larger models and supplied intermediate facts reduce the gap in shared-item comparisons; the data do not support "scale cannot help."

2. A limited training effect. Self-generated short-answer supervision improves original-task held-out scores. Context-conditioned targets outperform direct targets in six paired seeds, including an approximately accuracy-matched stage. The strongest claim is about this training procedure and evaluation setting.

3. Useful negative and limiting results. Confidence filtering and iterative refreshing do not show stable extra benefit. Longer training can reduce single-hop performance. The predefined overall MuSiQue transfer criterion is not met.

4. An unresolved explanation. Intermediate-token readability and target consistency are clues. Neither identifies a unique root cause. The effect could reflect several interacting factors: knowledge coverage, reference conventions, target semantics, confidence, and question-answer compatibility.

What a preprint should and should not claim

A defensible framing is *a controlled empirical study of self-generated supervision for compositional factual QA, including its gains and limits*. It should not claim a novel backpatching method, general autonomous self-improvement, a proven "hopping too late" mechanism, equal factual quality between matched targets, or demonstrated broad transfer.

Before manuscript submission, the key evidence gaps are independent factual adjudication of disputed examples, completion and frozen-protocol analysis of exp14, and a clear distinction between prospective tests and post hoc analyses. The report makes no claim that these unfinished items are already resolved.

Evidence record and selected references

How to read the statistics

All accuracies are percentages; differences are percentage points. Unless explicitly stated, ± is sample SD across training seeds. Reported 95% intervals are paired t intervals over three or six seeds on fixed items. They do not capture all item sampling, person/country clustering, target construction, reference truth, or dataset-selection uncertainty. A non-significant contrast is not evidence of equivalence. Repeated items and seeds are not independent new facts.

Source key and verification scope

Code	Local evidence used
E1 / E7	Revised gap comparison, common-item analysis and 120-item label audit; consolidated in the verified 11 October report.
E2	Original lens items plus saved overlap-filtered analysis; filtering was not rebuilt with a fresh tokenizer for this report.
E8 / E9 / E10	Saved exp8 summary, exp9 verified report, exp10 verified report; the full-self overlap is counted only once.
E11 / R11	Exp11 archived analysis and the 11 October consolidated report; includes raw-output rescoring verification and baseline reconciliation.
E12a / E12b	Job700454 review and job700972 seed extension; original scorer and six-seed outputs.
E12c	Job701713 v3 rescoring and offline review-flag sensitivity analysis; post hoc scoring revision preserved alongside original scores.
E12m	Job701792 matched785 local verification, independent analysis and target-label audit.
E13	Job701955 local verification: frozen MuSiQue data, scoring and adapter receipts.
D16 / P17	16 October offline consistency report; exp17 feasibility/design and preparation artifacts.
H	INWARD_HANDOFF, dated 3 October; historical numbers only where raw verification is missing.

This summary synthesizes saved analyses and their recorded checks; it does not claim a new independent audit of every raw prediction. Later verification notes report archive/code/data hashes and independent recomputation of scores and seed intervals. Recent downloads omit adapter weights and final scheduler accounting; completion relies on program markers and logs. Exact local paths and SHA-256 hashes for the inputs used here are preserved in the companion source_manifest.json. No original experimental outputs were overwritten, and no HPC experiments were launched.

Position in the literature

The gap measurement follows [1]; internal intervention is related to [2]. The self-target method is closely related to context distillation [3], applied here to internally generated intermediate facts and short-answer targets. MuSiQue [4] supplies the external composition benchmark. These connections motivate the study; this brief is not an exhaustive novelty review.

[1] Press, O. et al. (2023). Measuring and Narrowing the Compositionality Gap in Language Models. Findings of EMNLP.

[2] Biran, E. et al. (2024). Hopping Too Late: Exploring the Limitations of Large Language Models on Multi-Hop Queries. EMNLP.

[3] Snell, C., Klein, D. and Zhong, R. (2022). Learning by Distilling Context.

[4] Trivedi, H. et al. (2022). MuSiQue: Multihop Questions via Single-hop Question Composition. TACL 10:539-554.